

Olist E-Commerce Marketplace Analysis

Power BI Dashboard and SQL Validation Report

Prepared by: **Shubham Kumar Dubey**

Tools: Power BI, Power Query, DAX, Google Colab, SQLite SQL

Executive KPI Summary

Metric	Value
Total Revenue	13,591,643.70
Total Orders	99,441
Total Customers	96,096
Total Sellers	3,095
Total Products	32,951
Average Order Value	136.68
Average Review Score	4.09

Executive Summary

This project analyzes the Brazilian Olist E-Commerce marketplace dataset to understand revenue performance, customer geography, delivery efficiency, product category contribution, payment behavior, review patterns, and seller performance. The final solution includes a three-page Power BI dashboard and a SQL analysis notebook used to validate the major dashboard metrics.

- The marketplace generated total product revenue of 13.59M across 99,441 orders.
- Revenue was strongest in 2018, with 7.39M revenue from 54,011 orders.
- SP was the largest customer revenue state and also the strongest seller hub.
- Health Beauty generated the highest category revenue, while Bed Bath Table had the highest order volume.
- Credit card payments dominated the marketplace, contributing 78.34% of payment value.
- Delivery performance had a clear impact on satisfaction: on-time/early orders averaged 4.29 review score, while delayed orders averaged 2.27.

1. Project Objective

The objective of this project was to build a complete business intelligence analysis for an e-commerce marketplace. The dashboard was designed to help users monitor business growth, sales trends, delivery performance, product and seller contribution, and customer satisfaction.

2. Dataset and Tools

Area	Details
Dataset	Brazilian Olist E-Commerce public dataset
Data tables used	Customers, Orders, Order Items, Payments, Reviews, Products, Sellers, Category Translation
Main dashboard tool	Power BI
Data preparation	Power Query
Calculations	DAX measures and calculated columns
SQL validation	SQLite SQL in Google Colab
Output files	PBIX dashboard, SQL notebook, screenshots, PDF/DOCX report

3. Data Preparation and Model Design

The dataset was imported into Power BI and cleaned using Power Query. The category translation table was used to convert product category names into English, and custom columns were created for delivery analysis.

- Created Purchase Date, Delivery Days, Delay Days, Delivery Status, Order Year, and Month fields in the Orders table.
- Created a clean Category Name field by replacing underscores and formatting category names.
- Used Seller Short ID for readable seller visuals instead of full seller identifiers.
- Built relationships between customers, orders, order items, payments, reviews, products, sellers, and category translation tables.
- Used DAX measures for revenue, orders, customers, sellers, products, average order value, average review score, delivery metrics, and delay rate.

4. Dashboard Pages

The Power BI report contains three analytical pages: Executive Overview, Delivery and Review Analysis, and Product and Seller Analysis.

4.1 Executive Overview

This page presents the high-level business performance using KPI cards, yearly and monthly revenue trends, revenue by product category, revenue by customer state, delivery status, and payment method contribution.

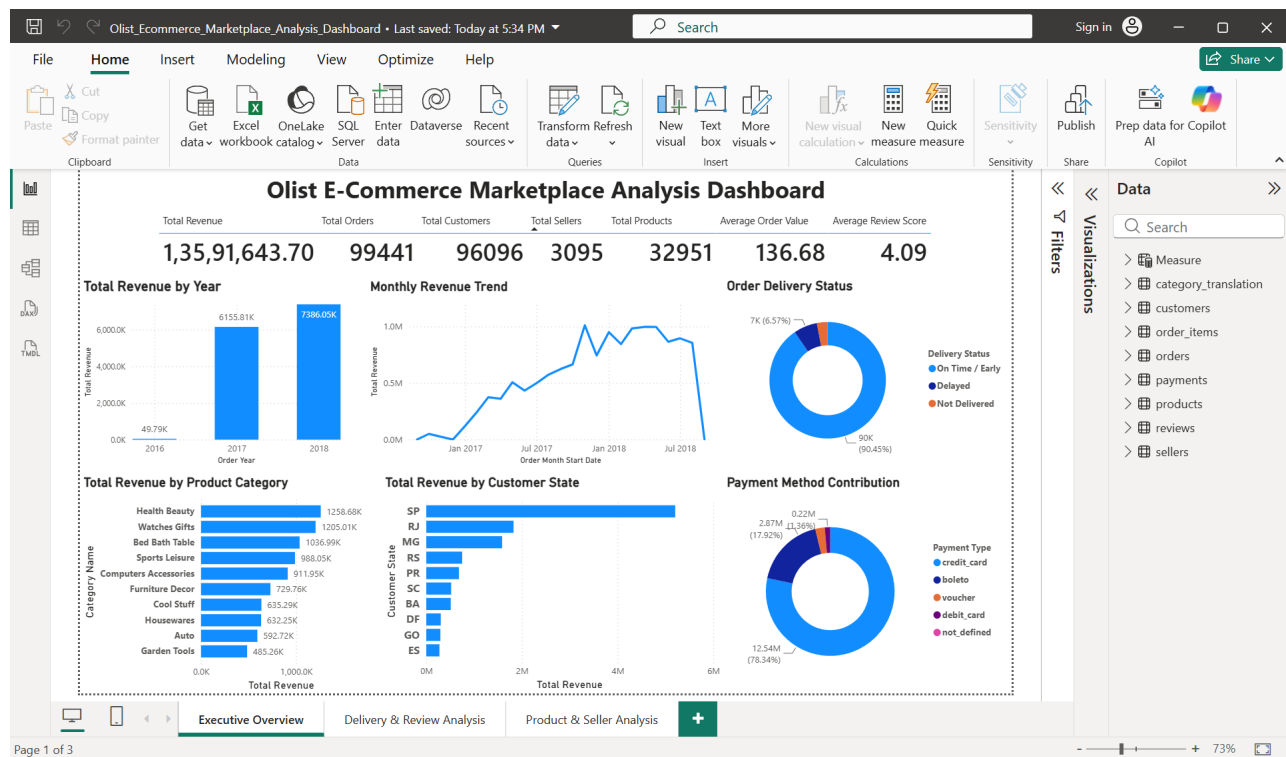


Figure 1: Executive Overview dashboard page

4.2 Delivery and Review Analysis

This page analyzes delivery speed, delay rate, delayed order count, review score distribution, and the relationship between delivery status and customer satisfaction.

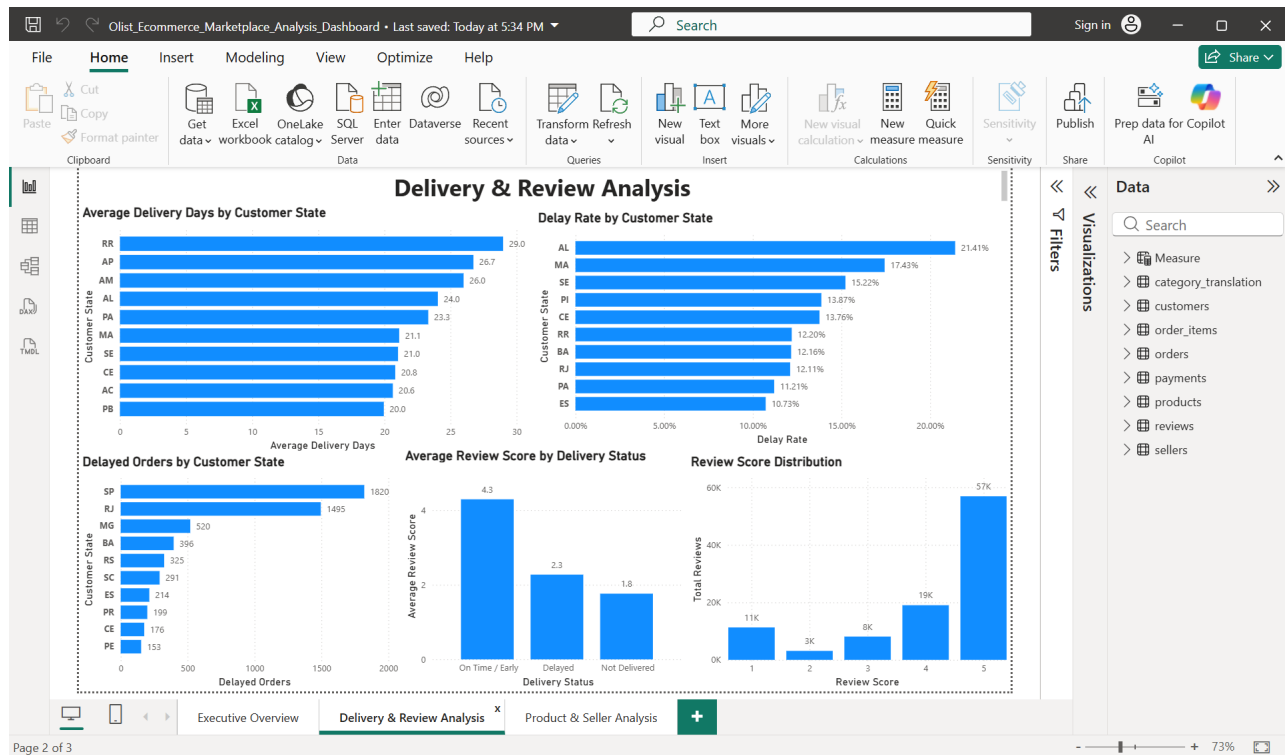


Figure 2: Delivery and Review Analysis dashboard page

4.3 Product and Seller Analysis

This page focuses on top sellers, high-performing product categories, seller state contribution, and product category order volume.

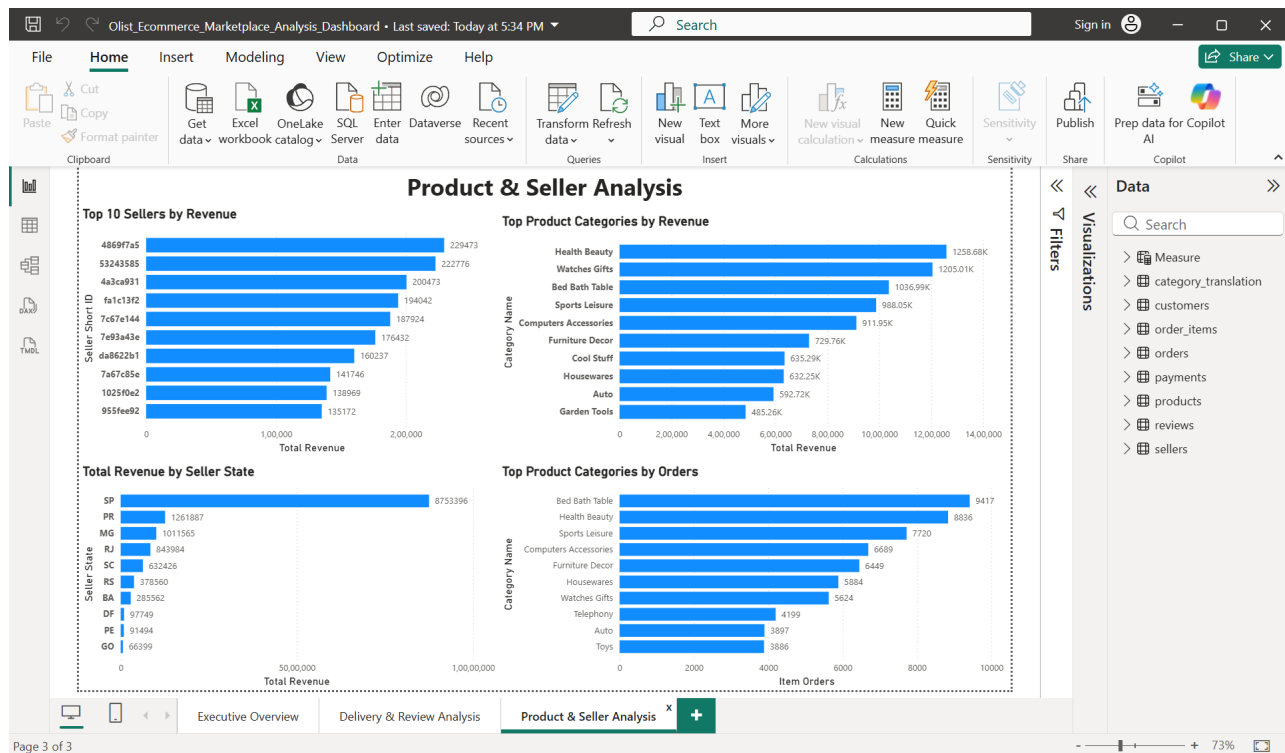


Figure 3: Product and Seller Analysis dashboard page

5. SQL Analysis and Validation

A SQL notebook was created in Google Colab using SQLite. Python was used only to load the Excel files into SQLite tables; the analysis itself was performed using SQL queries. The SQL outputs were used to validate the Power BI dashboard results.

No.	SQL Analysis Area	Purpose
1	Overall KPIs	Validate revenue, orders, customers, sellers, products, and AOV
2	Revenue by Year	Validate yearly revenue and order trends
3	Monthly Revenue Trend	Analyze month-wise order and revenue movement
4	Top Product Categories by Revenue	Identify highest revenue-generating categories
5	Revenue by Customer State	Find strongest customer markets by state
6	Payment Method Contribution	Measure payment value and percentage by method
7	Delivery Status Analysis	Classify orders as on-time/early, delayed, or not delivered
8	Delivery Performance by State	Analyze state-level delivery delay risk
9	Review Score by Delivery Status	Measure how delivery affects satisfaction
10	Review Score Distribution	Analyze customer review pattern
11	Top Sellers by Revenue	Identify top seller contribution
12	Revenue by Seller State	Analyze seller geography performance
13	Top Product Categories by Orders	Compare order volume by category
14	Delayed Orders by Customer State	Find states with the largest delayed order counts
15	Average Delivery Days by State	Find slowest delivery states

6. Key SQL-Validated Findings

6.1 Revenue and Growth

Year	Total Orders	Item Orders	Total Revenue
2016	329	312	49,785.92
2017	45,101	44,579	6,155,806.98
2018	54,011	53,775	7,386,050.80

Revenue grew strongly from 2017 to 2018. The highest monthly revenue periods included November 2017, April 2018, and May 2018.

6.2 Product Category Performance

Category	Orders	Items Sold	Revenue
Health Beauty	8,836	9,670	1,258,681.34
Watches Gifts	5,624	5,991	1,205,005.68
Bed Bath Table	9,417	11,115	1,036,988.68
Sports Leisure	7,720	8,641	988,048.97
Computers Accessories	6,689	7,827	911,954.32

Health Beauty was the top revenue category, while Bed Bath Table had the highest order and item volume. Watches Gifts generated high revenue despite lower order volume because of a higher average item price.

6.3 Customer Geography

Customer State	Orders	Customers	Revenue
SP	41,746	40,302	5,202,955.05
RJ	12,852	12,384	1,824,092.67
MG	11,635	11,259	1,585,308.03
RS	5,466	5,277	750,304.02
PR	5,045	4,882	683,083.76

SP was the leading customer state by revenue, orders, and customers. This indicates a high concentration of marketplace activity in Sao Paulo.

6.4 Payment Contribution

Payment Type	Orders	Payment Value	Contribution
Credit Card	76,505	12,542,084.19	78.34%
Boleto	19,784	2,869,361.27	17.92%
Voucher	3,866	379,436.87	2.37%
Debit Card	1,528	217,989.79	1.36%

Credit card was the dominant payment method, contributing more than three-fourths of total payment value.

6.5 Delivery and Customer Satisfaction

Delivery Status	Orders	Reviews	Avg Review Score
On Time / Early	89,941	89,308	4.29
Delayed	6,535	6,390	2.27
Not Delivered	2,965	2,848	1.76

The analysis shows a strong relationship between delivery performance and customer satisfaction. On-time or early orders received much higher review scores than delayed or not delivered orders.

6.6 Delay and Delivery Risk by State

State	Delivered Orders	Delayed Orders	Delay Rate
AL	397	85	21.41%
MA	717	125	17.43%
SE	335	51	15.22%
PI	476	66	13.87%
CE	1,279	176	13.76%

AL had the highest delay rate, while SP had the highest absolute delayed order count due to its very large order volume.

State	Total Orders	Delivered Orders	Avg Delivery Days
RR	46	41	28.98
AP	68	67	26.73
AM	148	145	25.99
AL	413	397	24.04
PA	975	946	23.32

6.7 Seller Performance

Seller Short ID	State	Orders	Items Sold	Revenue
4869f7a5	SP	1,132	1,156	229,472.63
53243585	BA	358	410	222,776.05
4a3ca931	SP	1,806	1,987	200,472.92
fa1c13f2	SP	585	586	194,042.03
7c67e144	SP	982	1,364	187,923.89

Most of the top revenue sellers were located in SP, confirming the dominance of Sao Paulo as a seller hub.

7. Business Recommendations

1. Focus logistics improvement on high-delay-rate states such as AL, MA, SE, PI, and CE.
2. Monitor high-volume states like SP and RJ because small delay improvements there can affect many customers.
3. Prioritize customer satisfaction initiatives for delayed and not delivered orders because review scores drop sharply in those groups.
4. Use Health Beauty and Watches Gifts as key revenue categories and Bed Bath Table as a volume-driving category.
5. Strengthen seller support and onboarding in states outside SP to reduce seller-side concentration risk.
6. Track payment behavior closely because credit cards dominate payment value, while boleto remains a meaningful secondary method.

8. Conclusion

The Olist E-Commerce Marketplace Analysis project successfully combines Power BI dashboard development with SQL validation. The analysis shows that revenue is concentrated in key states and categories, delivery performance has a major impact on customer reviews, and SP dominates both customer demand and seller supply. The project demonstrates practical skills in data cleaning, data modelling, DAX measure creation, SQL analysis, dashboard design, and business insight generation.

Appendix: Deliverables

File	Purpose
Olist_Ecommerce_Marketplace_Analysis_Dashboard.pbix	Power BI dashboard file
Olist_Ecommerce_SQL_Analysis.ipynb	SQL validation notebook
dashboard_page_1_executive_overview.png	Executive dashboard screenshot
dashboard_page_2_delivery_review_analysis.png	Delivery and review dashboard screenshot
dashboard_page_3_product_seller_analysis.png	Product and seller dashboard screenshot
Olist_Ecommerce_Project_Report.docx / .pdf	Final project report

Dataset source: [Brazilian E-Commerce Public Dataset by Olist on Kaggle](#)